

CS F407: Bayesian Networks with an LLM

AI Laboratory

TODO: Qwen code generation and comparison with the reference implementation.

Network and inference

Binary states are 0 = false and 1 = true. The graph has edges $C \rightarrow R$, $C \rightarrow S$, and $R, S \rightarrow W$, with factorisation

$$P(C, R, S, W) = P(C)P(R | C)P(S | C)P(W | R, S).$$

$P(C = 1) = 0.5$. Rain probabilities given $C = 0, 1$ are (0.2, 0.8); Sprinkler probabilities are (0.5, 0.1). For `evidence=["Rain", "Sprinkler"]`, WetGrass columns represent $(R, S) = (0, 0), (0, 1), (1, 0), (1, 1)$, with $P(W = 1)$ values (0.01, 0.90, 0.90, 0.99). Changing column order changes the model despite valid normalisation.

Query	Variable elimination	Enumeration
$P(R = 1 W = 1)$	0.704769231	0.704769231
$P(S = 1 W = 1)$	0.427846154	0.427846154
$P(C = 1 W = 1)$	0.574615385	0.574615385
$P(R = 1 W = 1, S = 0)$	0.992202205	0.992202205

Query variables are Rain, Sprinkler, Cloudy, and Rain. Evidence is WetGrass in the first three, plus Sprinkler=0 in the last. Wet grass increases Rain from its prior 0.5, Sprinkler from 0.3, and Cloudy from 0.5. With the sprinkler off, rain explains wet grass more strongly. All queries agree with independent enumeration of 16 assignments within 10^{-12} . Variable elimination multiplies factors, eliminates hidden variables by summation, and normalises.

Parameter estimation

The graph is fixed; data are simulated from the specified network. State names are explicitly fixed to $\{0, 1\}$. MLE was checked against conditional counts; BDeu was checked against the pseudo-count formula. Below, the sample-size experiment uses seed 7; the varying-seed experiment uses $N = 100$. Both estimate $P(R = 1 | C = 1)$.

N	MLE (seed 7)	Absolute error	Seed	MLE ($N = 100$)
20	0.888889	0.088889	1	0.693878
50	0.826087	0.026087	2	0.795455
100	0.708333	0.091667	3	0.744681
500	0.780488	0.019512	4	0.767857
1000	0.772541	0.027459	5	0.795918
5000	0.803961	0.003961	–	–

Method	$P(R = 1 C = 1)$
True parameter	0.800000
MLE, N=30, seed 11	0.909091
BDeu, N=30, seed 11, ESS=10	0.781250

For a binary node with q parent configurations, BDeu adds $10/(2q)$ to each state count. This explicit prior does not guarantee smaller error for every dataset.

Different samples produce different counts. Sampling variability explains these estimates; increasing sample sizes need not reduce every successive finite-sample error.

Validation and LLM exercises

Structure, normalisation, inference, MLE-count, and BDeu-count checks pass. A CPT with a column sum of 1.2 is rejected. A permuted WetGrass CPT passes `check_model()` but gives 0.717295227 for $P(R = 1 | W = 1)$ instead of 0.704769231. Local consistency does not establish correct probability semantics. The local code uses pgmpy 1.1.2 estimator instances `DiscreteMLE()` and `DiscreteBayesianEstimator()`. The installed `model.fit` API was inspected before execution. An LLM supplied local query code and explanations.

Reproduce. Run `../venv/bin/python solution.py` here. Results: `results.json`; prompt and correction record: `prompts.txt`.